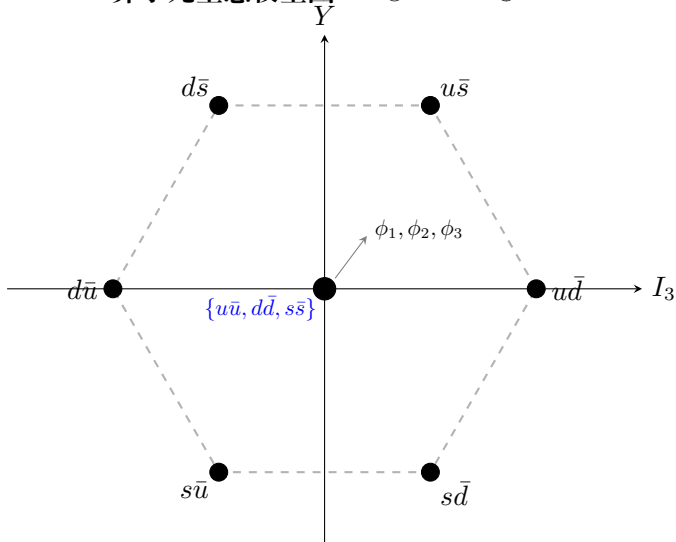


介子九重态权重图： $3 \otimes 3^* = 8 \oplus 1$



中心态波函数构造:

$$\text{八重态 } 8 \begin{cases} \phi_1 = \frac{1}{\sqrt{2}}(u\bar{u} - d\bar{d}) \\ \phi_2 = \frac{1}{\sqrt{6}}(u\bar{u} + d\bar{d} - 2s\bar{s}) \end{cases}$$

$$\text{单态 } 1 \quad \phi_3 = \frac{1}{\sqrt{3}}(u\bar{u} + d\bar{d} + s\bar{s})$$

注:

ϕ_1 为同位旋三重态分量 ($I = 1$);

ϕ_2, ϕ_3 为同位旋单态分量 ($I = 0$)。